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328311 (14)

BE (3rd Semester)

Examination, April-May, 2014

Branch : AEI, EI, Et & T

MATHEMATICS - III

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) is compulsory.

Solve any two from (b), (c) and (d).

Q. 1. (a) If x = c is a point of discontinuity then the Fourier series of f(x) at x = c gives f(x) equal to : 2

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(i) $\frac{1}{2}\{f(c-0)-f(c+0)\}$

(ii) $\frac{1}{2}+\{f(c-0)+f(c+0)\}$

(iii) $\frac{1}{2}f\{f(c+0)-f(c-0)\}$

(iv) None of these

(b) Find the Fourier series for : 7

$$f(x) \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

and find the value of $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\ldots$

(c) Find the Fourier series corresponding to the function f(x) defined in (–2, 2) as follows : 7

$$f(x)=\begin{cases} 2, & -2 \le x \le 0 \\ x, & 0 < x < 2 \end{cases}$$

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(d) Obtain the constant term and the coefficient of the first sine and cosine terms in the Fourier series f(x) as given in the following table : 7

x :	0	1	2	3	4	5
f(x) :	9	18	24	28	26	20

Q. 2. (a) If F(s) is the complex Fourier transform of f(x) and : 2

$$F_c\{f(ax)\} = K.F_c\left(s/a\right)$$

then value of K is

(i) 1

(ii) a

(iii) 1/a

(iv) None of these

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(b) Express $f(x) = \begin{cases} 1 & , \quad 0 \le x \le \pi \\ 0 & , \quad x > \pi \end{cases}$

as a fourier sine integral and hence evaluate : 7

$$\int_0^\infty \frac{1-\cos(\pi\lambda)}{\lambda} \sin(x\lambda)\, d\lambda$$

(c) Find the Fourier transform of : 7

$$f(x) = \begin{cases} 1 & , \quad |x| < 1 \\ 0 & , \quad |x| > 1 \end{cases}$$

Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.

(d) Find the Fourier cosine transform of e^{-x^2}. 7

Q. 3. (a) $L^{-1}\left\{\frac{1}{s^n}\right\}$ is possible only when n is : 2

(i) Zero

(ii) Negative integer

(iii) Positive integer

(iv) All of the above

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(b) (i) Find Laplace transform of $t^2 \cos at$ 3

(ii) Prove that $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt = \log 3$ 4

(c) (i) Find $L^{-1}\left\{\frac{1}{s(s^2+a^2)}\right\}$ 3

(ii) Find inverse Laplace transform of : 4

$\log\left(\frac{s+1}{s-1}\right)$

(d) Solve the given differential equation using Laplace transform : 7

$t.y'' + 2y' + ty = \cos t$

given that $y(0) = 1$

Q. 4. (a) If $f(z) = e^{2z}$, then the imaginary part of $f(z)$ is : 2

(i) $e^y \sin x$

(ii) $e^x \cos y$

(iii) $e^{2x} \cos 2y$

(iv) $e^{2x} \sin 2y$

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(b) If $u(x, y) = \frac{\sin 2x}{\cosh 2y + \cos 2x}$, construct the analytic function $f(z)$. 7

(c) Find the series expansion of the function : 7

$$f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$$

about $z = 0$, in the regions

(i) $|z| < 2$

(ii) $2 < |z| < 3$

(iii) $|z| > 3$

(d) Evaluate the integral : 7

$$\int_0^\pi \frac{1 + 2\cos\theta}{5 + 4\cos\theta} d\theta$$

using contour integral.

Q. 5. (a) The correlation coefficient is 0, the two regression lines are : 2

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(i) Parallel

(ii) Perpendicular

(iii) Coincident

(iv) Inclined at 45° to each other

(b) The students got the following percentage of marks in Mathematics and Mechanics : 7

Roll No.	1	2	3	4	5	6	7	8	9	10
Marks in Mathematics	78	36	98	25	75	82	90	62	65	39
Marks in Mechanics	84	51	91	60	68	62	86	58	53	47

Calculate the coefficient of correlation.

(c) Calculate the rank correlation coefficient from the following table : 7

x :	78	89	97	69	59	79	68	57
y :	125	137	156	112	107	136	123	108

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(d) In the following table are recorded data showing the test scores made by salesman on an intelligence test and their weekly sales : 7

Salesman	1	2	3	4	5	6	7	8	9	10
Test scores	40	70	50	60	80	50	90	40	60	60
Sales	2.5	6	4.5	5	4.5	2	5.5	3	4.5	3

Calculate the regression line of sales on test scores and estimate the most probable weekly sales volume if a salesman makes a score of 70.

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